1 **Design & Implementation of Human-Computer Interfaces**
NPTEL-MOOCS

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2 **L2 – Introduction to Usability**

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3 **Recap – last Lecture**

- We talked about the basic idea of interactive systems
- We also learned about the historical evolution of the area in brief
- Today, we shall talk about a key concern in designing such systems - usability

4 **Design Concerns**

5 **A Definition!**

"Human-computer interaction is a discipline concerned with the design, evaluation and implementation of interactive computing systems for human use and with the study of major phenomena surrounding them." (ACM SIGCHI)

6 **Core Concern**

- Incorporating humans into the design

7

Human Factors

8 **Role of Human Factors**

- We wish to design for the "laymen" users
  - Who are they?
  - Why to bother about such "characterization"?

9 **Understanding Human Factors**

- Let's try to understand the nature of the users and its importance with an example

10 **Example**

- Consider someone who wants to perform railway ticket booking (traveler/agent)

11 **Example**

- What the user is expected to do?

12 **Example**
- Input station names
- Check availability
- Book tickets
- Make payments

13 **Example**
- Now consider a DBMS manager who performs database querying – to get the same information from the Railway database

14 **Example**
- Is this user performing *totally different* things than the earlier one?

15 **Example**
- Technically, both are performing the same work

16 **Example**
- There is the railway database
- Both queries the database
- Both retrieves the information
- Both uses the information

17 **Example**
- Difference in "expectations" and "need" of the users

18 **User Classification**
- We need to "know the users" in order to design

19 **User Classification**
- Generic classification of users
  - Novice
  - Intermittent
  - Expert

20 **User Classification**
- Example - saving a file with "Ctrl+S" vis-à-vis through menu option in MS Word

21 **Interdisciplinary**
- Identifying human factors and incorporate those in the design requires knowledge and expertise in many field of studies

22 **Which Discipline?**

23

Considering Human in Design

24 **Different Design Perspectives**

- The same term (User-Centric Design) can be viewed by different stakeholders in different ways

25 **Different Design Perspectives**

- A *designer* is primarily concerned about the design of the interface elements and layouts – the creative design aspects

26 **Taking Humans into the Design**

- For a *(industrial) product designer*, it might refer to the *form* (shape, size and look) of the product

27 **Different Design Perspectives**

- An *electronic engineer* might look at it as adding more features at the hardware level, such as more sensors, smaller chips to reduce size or more power efficient battery

28 **Different Design Perspectives**

- We are mostly concerned about the perspective of the *application software developer*, who needs to follow a development life cycle that takes care of the user characteristics, so as to better match the users' needs and expectations

29 **Usability**

30 **Usability**

- We require an explicit measure to judge our design – if we have taken care of the human factors

The measure is called "Usability"

31 **Usability: Definitions and Standards**

- ISO definition (ISO 9241-210:2009) of usability - "the extent to which a product can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use."

32 **Usability: Definitions and Standards**

- Three crucial aspects in the definition

33 **Usability: Definitions and Standards**

- The product is meant to be used by a *specified* group of users
- In other words, a usable product need not be designed "for all"

34 **Usability: Definitions and Standards**

- The product should allow the users to achieve *specified* set of goals
- Thus, putting every conceivable features in a product not necessarily leads to a

usable product

35 **Usability: Definitions and Standards**

- The product should be designed for *specified* context of usage
- Clearly, a usable product need not be so for "all" usage scenario

36 **Usability: Definitions and Standards**

- Definition reveals THREE measures
  - Effectiveness
  - Efficiency
  - Satisfaction

37 **Usability: Definitions and Standards**

- Jacob Nielsen [2012] argued that usability alone cannot make a product "useful"

38 **Usability: Definitions and Standards**

- An acceptable product should have two quality attributes
- One is usability and the other is "utility"

39 **Usability: Definitions and Standards**

- Nielsen proposed FIVE quality components of usability

40 **Usability: Definitions and Standards**

- Learnability: the "ease" with which a first time (novice) user performs "basic" tasks with the system

41 **Usability: Definitions and Standards**

- Efficiency: The speed at which the users can complete tasks

42 **Usability: Definitions and Standards**

- Memorability: The "ease" with which an intermittent user, who returns to use the system occasionally (after some gaps), can reestablish "proficiency"

43 **Usability: Definitions and Standards**

- Errors: The rate at which the users make errors, the "severity" of those errors and "ease" with which the users can recover from errors

44 **Usability: Definitions and Standards**

- Satisfaction: How pleasant is it to use the design

45 **Usability: Definitions and Standards**

- Utility refers to the "functionality" that the design is supposed to serve
  - Measure of the extent to which design supports the "functional needs" (the features) of the users

46 **Usability: Definitions and Standards**

- If we compare Nielsen's framework to the ISO definition, we observe "effectiveness" might be mapped to "utility"

47 **Usability: Definitions and Standards**

- The ISO definition therefore provided only two measures for usability: the efficiency and satisfaction

48 **Usability: Definitions and Standards**

- The five components of usability offer a more precise measure - we shall make use of these components in the subsequent lectures

49 **User-Centered Design**

50 **User-Centered Design**

- Term coined by Shneiderman (1986)
- Objective - to design products that increase usability

51 **User-Centered Design**

- Indicates active/passive involvement of users in the design life cycle

52 **User-Centered Design**

- Related terms
  - "Cooperative design" [Greenbaum and Kyng, 1992]
  - "Participatory design" [Schular and Namioka, 1993]
  - "Contextual design" [Beyer and Holtzblatt, 1997]

53 **User-Centered Design**

- ISO in its standards prefers to use the term "human-centered design"

54 **Book**

- Bhattacharya, S. (July, 2019). Human-Computer Interaction: User-Centric Computing for Design, McGraw-Hill India
  - Print Edition: ISBN-13: 978-93-5316-804-9; ISBN-10: 93-5316-804-X

• E-book Edition: ISBN-13: 978-93-5316-805-6; ISBN-10: 93-5316-805-8

Chapter 2, Sec 2.1 – 2.4.2